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Network Monitoring and Traffic Analysis Lab

A Cybersecurity Assessment Project

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NETWORKING PROJECT

INTRODUCTION

Network monitoring and traffic analysis are essential components of cybersecurity and network administration. Understanding how devices communicate across a network enables security professionals to identify normal traffic patterns, detect anomalies, and assess potential security risks.

This project was conducted to gain practical experience in network traffic monitoring, packet analysis, host discovery, service enumeration, operating system fingerprinting, attack surface assessment, and cybersecurity automation. Industry-standard tools including Wireshark, Nmap, Python, Scapy, and Matplotlib were utilized to analyze network communications, identify active services, and automate traffic analysis tasks.

The project follows a structured approach beginning with networking fundamentals and progressing toward advanced security assessment and automation techniques commonly used by security analysts and penetration testers.

Tools and Technologies Used

Tool	Purpose
Wireshark	Live packet capture and deep protocol inspection
Nmap	Host discovery, port scanning, OS and service fingerprinting
Python / Scapy	Automated pcap analysis and protocol classification
Matplotlib	Data visualisation – protocol distribution and host activity charts
Windows CLI	ipconfig, nslookup, ping, tracert, arp for baseline network analysis

UNDERSTANDING HOW NETWORKS WORK

```
PS C:\Users\user> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::aa8a:e40:24db:93ff%
19
    Autoconfiguration IPv4 Address. . . : 169.254.75.247
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::ec83:526a:94b4:7fc%
15
    IPv4 Address. . . . . : 192.168.159.13
```

Figure 1: Network configuration information collected using the Windows ipconfig utility, displaying active network interfaces, IPv4 addressing, subnet mask configuration, and local network connectivity details.

```
    Link-local IPv6 Address . . . . . : fe80::ec83:526a:94b4:7fc%
15
    IPv4 Address. . . . . : 192.168.159.13
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.159.121
PS C:\Users\user> nslookup google.com
Server: Unknown
Address: 192.168.159.121

Non-authoritative answer:
Name:   google.com
Addresses:  2404:6800:4009:819::200e
           142.250.195.78

PS C:\Users\user> ping google.com

Pinging google.com [142.250.195.78] with 32 bytes of data:
Reply from 142.250.195.78: bytes=32 time=142ms TTL=114
Reply from 142.250.195.78: bytes=32 time=72ms TTL=115
Reply from 142.250.195.78: bytes=32 time=108ms TTL=115
Reply from 142.250.195.78: bytes=32 time=93ms TTL=115

Ping statistics for 142.250.195.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 72ms, Maximum = 142ms, Average = 103ms
PS C:\Users\user> ping google.com
```

Figure 2: DNS resolution and connectivity verification using nslookup and ping. The results show successful domain name resolution for google.com and successful ICMP Echo Request/Reply communication with no packet loss.

```

Pinging google.com [142.250.195.78] with 32 bytes of data:
Reply from 142.250.195.78: bytes=32 time=161ms TTL=115
Reply from 142.250.195.78: bytes=32 time=73ms TTL=115
Reply from 142.250.195.78: bytes=32 time=63ms TTL=115
Reply from 142.250.195.78: bytes=32 time=73ms TTL=115

Ping statistics for 142.250.195.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 63ms, Maximum = 161ms, Average = 92ms
PS C:\Users\user> tracert google.com

Tracing route to google.com [142.250.195.78]
over a maximum of 30 hops:

  0  0 ms  0 ms  0 ms  192.168.159.121
  1  *      *      *      Request timed out.
  2  51 ms  33 ms  87 ms  10.174.203.174
  3  *      *      *      Request timed out.
  4  72 ms  28 ms  45 ms  223.224.19.18
  5  *      45 ms  *      125.214.191.37
  6  *      40 ms  45 ms  125.214.191.36
  7  57 ms  44 ms  54 ms  125.214.162.103
  8  404 ms  62 ms  78 ms  ge-2-0-0.mala.aggr-01.dialog.lk [
125.214.162.70]
  9  112 ms  83 ms  69 ms  142.250.172.106
 10  287 ms  86 ms  61 ms  192.178.110.123
 11  85 ms  94 ms  81 ms  142.251.69.45
 12  74 ms  69 ms  74 ms  lcboma-be-in-f14.1e100.net [142.
250.195.78]

```

Trace complete.

Figure 3: Network connectivity testing performed using ICMP ping requests to google.com, showing successful packet transmission, round-trip times, and zero packet loss.

```

PS C:\Users\user> arp -a

Interface: 192.168.159.13 --- 0xf
    Internet Address      Physical Address      Type
    -----
    192.168.159.121       5e-9e-14-37-cc-66    dynamic
    192.168.159.255       ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251          01-00-5e-00-00-fb    static
    224.0.0.252          01-00-5e-00-00-fc    static
    239.255.255.250      01-00-5e-7f-ff-fa    static
    255.255.255.255      ff-ff-ff-ff-ff-ff    static

Interface: 169.254.75.247 --- 0x13
    Internet Address      Physical Address      Type
    -----
    169.254.255.255      ff-ff-ff-ff-ff-ff    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251          01-00-5e-00-00-fb    static
    224.0.0.252          01-00-5e-00-00-fc    static
    239.255.255.250      01-00-5e-7f-ff-fa    static
    255.255.255.255      ff-ff-ff-ff-ff-ff    static
PS C:\Users\user>

```

Figure 4: Address Resolution Protocol (ARP) table obtained using the arp -a command, displaying IP-to-MAC address mappings and illustrating address resolution within the local network.

The first phase of the project focused on understanding fundamental networking concepts and analyzing the local network environment. Network configuration information was collected using Windows networking utilities, including ipconfig, nslookup, ping, tracert, and arp. The analysis identified the device's IPv4 address, subnet mask, default gateway, and DNS configuration. DNS resolution was tested by resolving the domain name google.com to its corresponding IP address. Network connectivity and latency were evaluated using ICMP ping requests, while traceroute was used to examine the packet path

between the local host and Google's servers. Additionally, ARP table analysis was performed to observe the mapping between IP addresses and MAC addresses within the local network. These activities provided a practical understanding of IP addressing, DNS resolution, routing, network communication, and address resolution mechanisms, forming the foundation for subsequent packet analysis and network monitoring tasks.

PACKET CAPTURE AND TRAFFIC ANALYSIS USING WIRESHARK

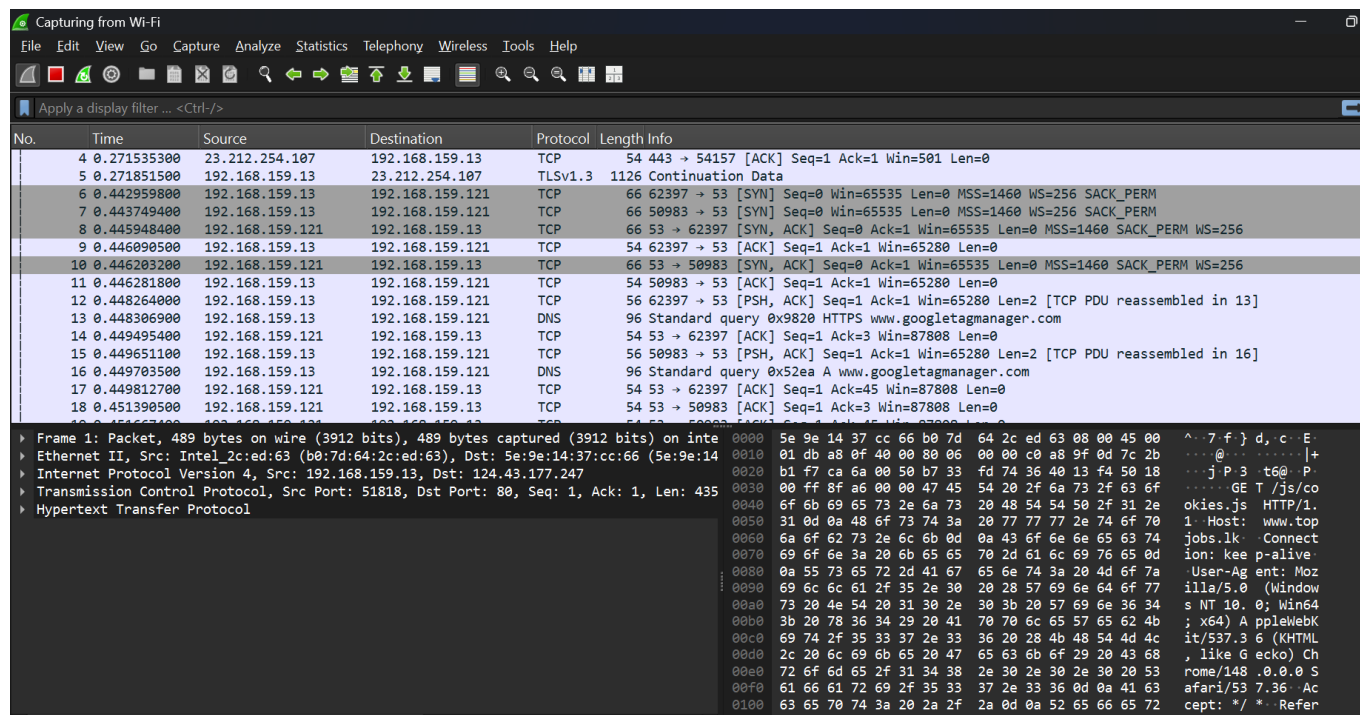


Figure 5: Wireshark packet capture displaying TCP and HTTP communications between the local host and external servers. The analysis provides visibility into source and destination IP addresses, transport-layer connections, packet details, and application-layer traffic exchanged during network communication.

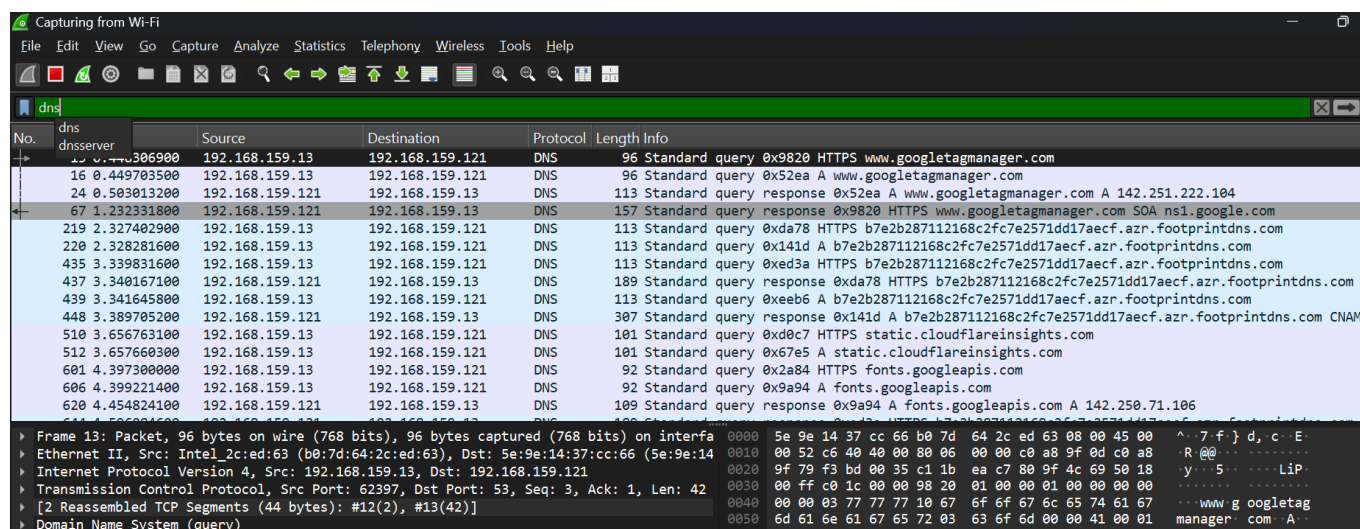


Figure 6: Wireshark DNS traffic analysis displaying DNS queries and responses generated during web browsing activities. The capture illustrates the domain name resolution process, including requests for domain records and corresponding responses from the DNS server.


```

Windows PowerShell
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PS C:\Users\user> nslookup google.com
Server: UnKnown
Address: 192.168.159.121

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4009:826::200e
          142.250.207.238

PS C:\Users\user> |

```

Figure 7: Domain Name System (DNS) lookup performed using the nslookup utility. The results demonstrate successful domain name resolution for google.com, returning both IPv4 and IPv6 addresses provided by the configured DNS server.

No.	Time	Source	Destination	Protocol	Length	Info
6	0.442959800	192.168.159.13	192.168.159.121	TCP	66	62397 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
7	0.443749400	192.168.159.13	192.168.159.121	TCP	66	50983 → 53 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
8	0.445948400	192.168.159.121	192.168.159.13	TCP	66	53 → 62397 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM WS=256
9	0.446090500	192.168.159.13	192.168.159.121	TCP	54	62397 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
10	0.446283200	192.168.159.121	192.168.159.13	TCP	66	53 → 50983 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 SACK_PERM WS=256
11	0.446281800	192.168.159.13	192.168.159.121	TCP	54	50983 → 53 [ACK] Seq=1 Ack=1 Win=65280 Len=0
12	0.448264000	192.168.159.13	192.168.159.121	TCP	56	62397 → 53 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=2 [TCP PDU reassembled in 13]
13	0.448306900	192.168.159.13	192.168.159.121	DNS	96	Standard query 0x9820 HTTPS www.googletagmanager.com
14	0.449495400	192.168.159.121	192.168.159.13	TCP	54	53 → 62397 [ACK] Seq=1 Ack=3 Win=87808 Len=0
15	0.449651100	192.168.159.13	192.168.159.121	TCP	56	50983 → 53 [PSH, ACK] Seq=1 Ack=1 Win=65280 Len=2 [TCP PDU reassembled in 16]
16	0.449703500	192.168.159.13	192.168.159.121	DNS	96	Standard query 0x52ea A www.googletagmanager.com
17	0.449812700	192.168.159.121	192.168.159.13	TCP	54	53 → 62397 [ACK] Seq=1 Ack=45 Win=87808 Len=0
18	0.451390500	192.168.159.121	192.168.159.13	TCP	54	53 → 50983 [ACK] Seq=1 Ack=3 Win=87808 Len=0
19	0.451667400	192.168.159.121	192.168.159.13	TCP	54	53 → 50983 [ACK] Seq=1 Ack=45 Win=87808 Len=0
20	0.451667800	192.168.159.121	192.168.159.13	TCP	55	53 → 50983 [PSH, ACK] Seq=1 Ack=45 Win=87808 Len=1 [TCP PDU reassembled in 24]

Figure 8: Wireshark capture displaying TCP communication and the TCP three-way handshake process. The analysis illustrates SYN, SYN-ACK, and ACK packets exchanged between the client and server during connection establishment, demonstrating the operation of reliable transport-layer communications.

```

Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\user> ping google.com

Pinging google.com [142.250.207.142] with 32 bytes of data:
Reply from 142.250.207.142: bytes=32 time=56ms TTL=115
Reply from 142.250.207.142: bytes=32 time=65ms TTL=115
Reply from 142.250.207.142: bytes=32 time=75ms TTL=115
Reply from 142.250.207.142: bytes=32 time=72ms TTL=115

Ping statistics for 142.250.207.142:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 56ms, Maximum = 75ms, Average = 67ms
PS C:\Users\user> |

```

Figure 9: ICMP connectivity analysis using the ping utility. The results show successful communication with google.com, including round-trip time measurements, packet delivery statistics, and zero packet loss, confirming network connectivity and performance.

The image shows a Wireshark packet capture of ICMP traffic. The packet list pane displays several ICMP Echo (ping) request and reply packets. The packet details pane shows the structure of an ICMP Echo (ping) request, including the type, code, checksum, and sequence number. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and Internet Control Message Protocol header.

No.	icmp	Source	Destination	Protocol	Length	Info
748	25.241763200	192.168.159.13	142.250.207.142	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 748)
749	25.241763200	142.250.207.142	192.168.159.13	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=115 (request in 747)
764	26.194553000	192.168.159.13	142.250.207.142	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 765)
765	26.194553000	142.250.207.142	192.168.159.13	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=115 (request in 764)
766	27.224348100	192.168.159.13	142.250.207.142	ICMP	74	Echo (ping) request id=0x0001, seq=3/768, ttl=128 (reply in 767)
767	27.224348100	142.250.207.142	192.168.159.13	ICMP	74	Echo (ping) reply id=0x0001, seq=3/768, ttl=115 (request in 766)
768	28.240916200	192.168.159.13	142.250.207.142	ICMP	74	Echo (ping) request id=0x0001, seq=4/1024, ttl=128 (reply in 769)
769	28.240916200	142.250.207.142	192.168.159.13	ICMP	74	Echo (ping) reply id=0x0001, seq=4/1024, ttl=115 (request in 768)

Figure 10: Wireshark capture displaying ICMP Echo Request and Echo Reply packets generated during a ping operation. The analysis illustrates the exchange of ICMP messages used to verify network connectivity, measure latency, and confirm successful communication between the source and destination hosts.

The image shows a Wireshark packet capture of UDP and QUIC traffic. The packet list pane displays several UDP and QUIC packets. The packet details pane shows the structure of a UDP packet, including the source and destination ports, length, and checksum. The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and User Datagram Protocol header.

No.	udp	Source	Destination	Protocol	Length	Info
1292	51069 → 443	192.168.159.13	142.250.134.95	UDP	1292	Len=1250
1293	Initial, DCID=bd9b914caab32907, PKN: 1, PADDING, PING, PING, CRYPTO, PADDING, CRYPTO, CRY...	192.168.159.13	142.250.134.95	QUIC	1292	Initial, DCID=bd9b914caab32907, PKN: 2, CRYPTO
1294	Initial, DCID=bd9b914caab32907, PKN: 3, CRYPTO, PADDING, CRYPTO, CRYPTO, PADDING, CRYPTO, PADDI...	192.168.159.13	142.250.134.95	QUIC	1292	Initial, DCID=bd9b914caab32907, PKN: 3, CRYPTO, PADDING, CRYPTO, CRYPTO, PADDING, CRYPTO, PADDI...
1295	0-RTT, DCID=bd9b914caab32907	192.168.159.13	142.250.134.95	QUIC	123	0-RTT, DCID=bd9b914caab32907
1296	0-RTT, DCID=bd9b914caab32907	192.168.159.13	142.250.134.95	QUIC	661	0-RTT, DCID=bd9b914caab32907
1297	51069 → 443	192.168.159.13	142.250.134.95	UDP	1292	51069 → 443 Len=1250
1298	443 → 51069	142.250.134.95	192.168.159.13	UDP	1292	443 → 51069 Len=1250
1299	51069 → 443	192.168.159.13	142.250.134.95	UDP	78	51069 → 443 Len=36
1300	51069 → 443	192.168.159.13	142.250.134.95	UDP	71	51069 → 443 Len=29
1301	51069 → 443	192.168.159.13	142.250.134.95	UDP	71	51069 → 443 Len=29
1302	56904 → 443	192.168.159.13	142.250.134.95	UDP	1292	56904 → 443 Len=1250
1303	443 → 51069	142.250.134.95	192.168.159.13	UDP	75	443 → 51069 Len=33
1304	443 → 51069	142.250.134.95	192.168.159.13	UDP	103	443 → 51069 Len=61
1305	51069 → 443	192.168.159.13	142.250.134.95	UDP	182	51069 → 443 Len=140

Figure 11: UDP and QUIC traffic analysis in Wireshark, showing connectionless communication between the local workstation and external web servers during web browsing activities.

Wireshark was used to capture and analyze live network traffic generated through web browsing, DNS resolution, and connectivity testing activities. DNS packets were inspected to observe domain name resolution processes, including A and AAAA record lookups. TCP traffic analysis identified the three-way handshake mechanism used to establish reliable connections, while UDP traffic associated with modern web applications and QUIC communications was also observed. ICMP Echo Request and Echo Reply packets generated through ping operations were analyzed to evaluate network connectivity, latency, and packet delivery. Packet inspection provided visibility into source and destination addresses, protocol behavior, and communication patterns, strengthening practical knowledge of network traffic analysis and cybersecurity monitoring techniques.

NETWORK DISCOVERY AND PORT SCANNING WITH NMAP

```
PS C:\Users\user> nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: i686-pc-windows-windows
Compiled with: nmap-liblua-5.4.8 openssl-3.0.16 nmap-libssh2-1.1
1.1 nmap-libz-1.3.2 nmap-libpcap-1.0.47 Npcap-1.88 nmap-libdnet-
1.18.0 ipv6
Compiled without:
Available nsock engines: iocp poll select
PS C:\Users\user> nmap -sn 192.168.159.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-01 17:06 +053
0
Nmap scan report for 192.168.159.121
Host is up (0.0031s latency).
MAC Address: 5E:9E:14:37:CC:66 (Unknown)
Nmap scan report for 192.168.159.13
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 12.10 second
```

Figure 12: Nmap host discovery scan performed on the local subnet (192.168.159.0/24), identifying active devices connected to the network.

```
PS C:\Users\user> nmap 192.168.159.121
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-01 17:06 +053
0
Nmap scan report for 192.168.159.121
Host is up (0.0025s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 5E:9E:14:37:CC:66 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 22.53 seconds
PS C:\Users\user> nmap -sV 192.168.159.121
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-01 17:08 +053
0
Nmap scan report for 192.168.159.121
Host is up (0.0022s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  dnsmasq 2.51
MAC Address: 5E:9E:14:37:CC:66 (Unknown)

Service detection performed. Please report any incorrect results
at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 29.00 seconds
PS C:\Users\user>
```

Figure 13: Nmap port scanning and service enumeration results identifying TCP port 53 as open and detecting the dnsmasq 2.51 DNS service running on the gateway device.

Nmap was used to perform network reconnaissance and identify active hosts within the local subnet. A host discovery scan revealed two active devices, including the local workstation and the network gateway. Subsequent port scanning of the gateway device identified TCP port 53 as open. Service enumeration was

then performed using version detection techniques, revealing that the device was running the dnsmasq 2.51 service, which provides DNS functionality within the local network. This exercise demonstrated how network services can be identified and assessed through reconnaissance techniques, providing valuable insight into network exposure and attack surface analysis.

Understanding Port Scan Mechanics

Wireshark captures taken during Nmap scans illustrated the TCP-based probing process:

- **Open port:** target responds with SYN-ACK
- **Closed port:** target returns RST-ACK

This exercise demonstrated how defenders can identify scanning behaviour through traffic analysis – repeated SYN packets across sequential ports are a reliable indicator of reconnaissance activity.

UNDERSTANDING HOW PORT SCANS WORK (ATTACKER VS DEFENDER)

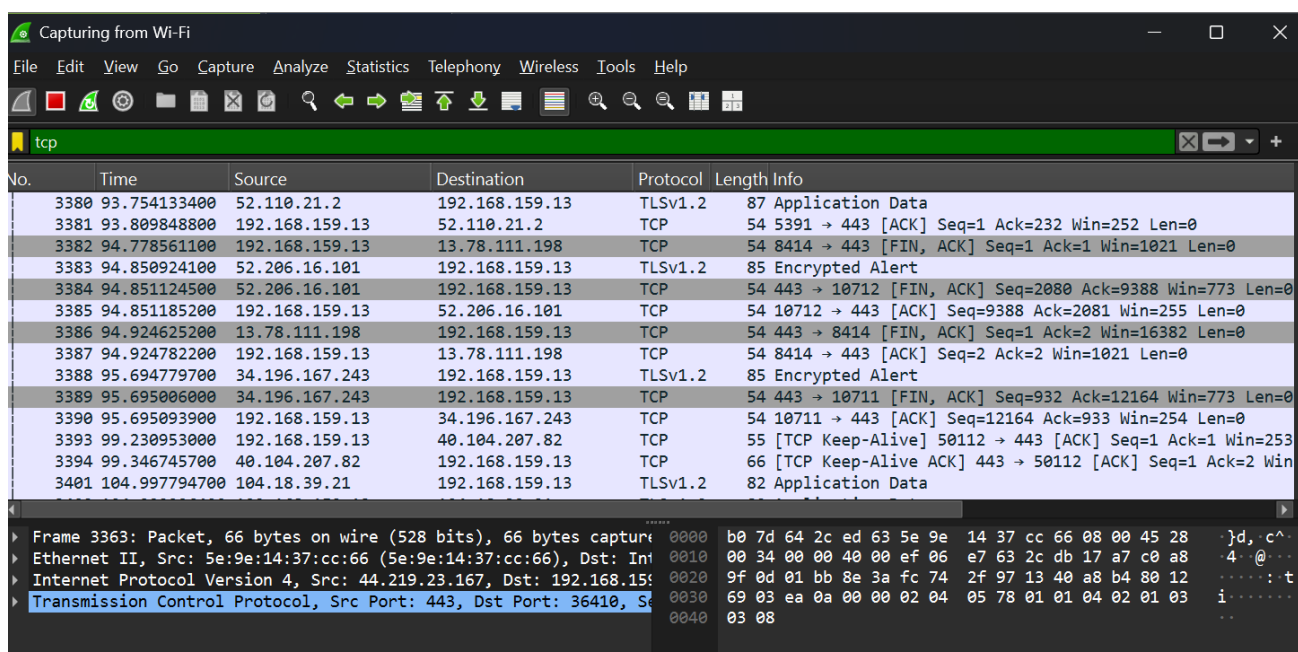


Figure 14: TCP packet analysis in Wireshark showing SYN, ACK, FIN, and TCP Keep-Alive packets exchanged between the local workstation and remote hosts during network communication.

```

Nmap done: 1 IP address (1 host up) scanned in 22.53 seconds
PS C:\Users\user> nmap -sV 192.168.159.121
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-01 17:08 +0530
Nmap scan report for 192.168.159.121
Host is up (0.0022s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  dnsmasq 2.51
MAC Address: 5E:9E:14:37:CC:66 (Unknown)

Service detection performed. Please report any incorrect results
at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 29.00 seconds

```

Figure 15: Nmap service version detection results identifying TCP port 53 as open and confirming the presence of the dnsmasq 2.51 DNS service on the gateway device (192.168.159.121).

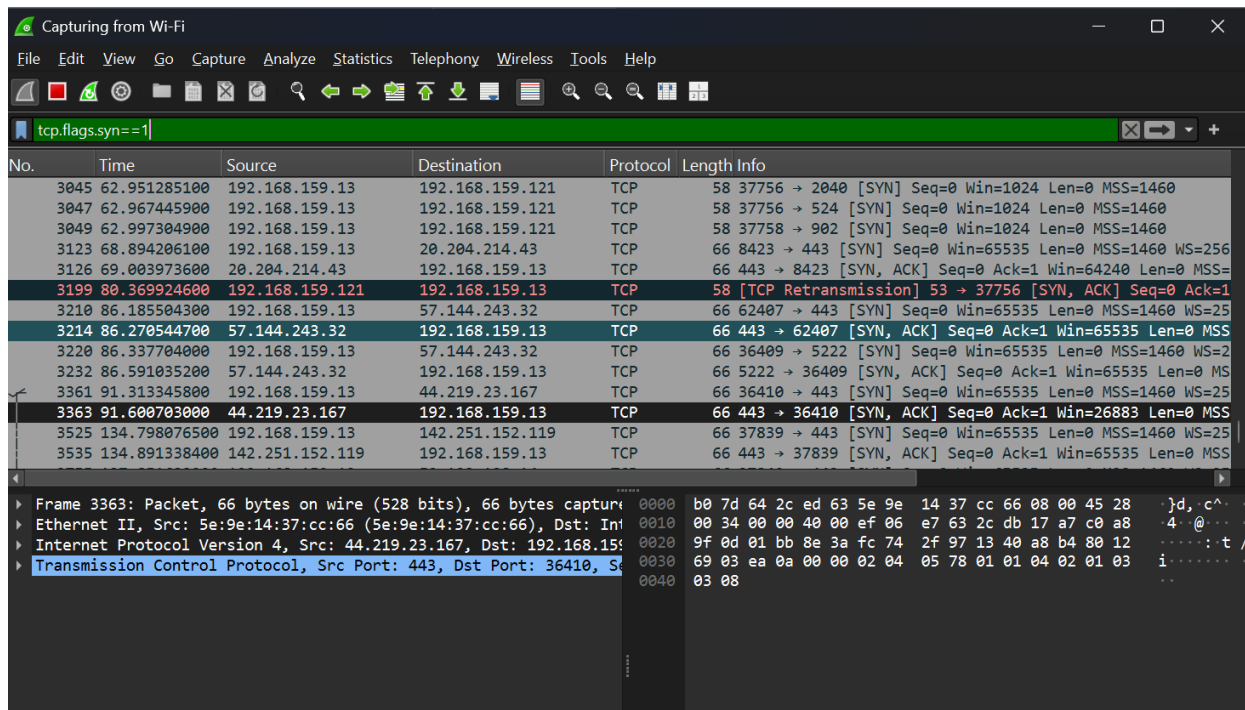


Figure 16: Wireshark capture displaying TCP SYN and SYN-ACK packets exchanged during network scanning and connection establishment activities, illustrating the operation of TCP-based reconnaissance techniques.

```

PS C:\Users\user> nmap 192.168.159.121
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-01 17:15 +0530
Nmap scan report for 192.168.159.121
Host is up (0.0026s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 5E:9E:14:37:CC:66 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 22.25 seconds
PS C:\Users\user>

```

Figure 17: Nmap port scan results identifying TCP port 53 as open on the gateway device (192.168.159.121), indicating the presence of an active DNS service.

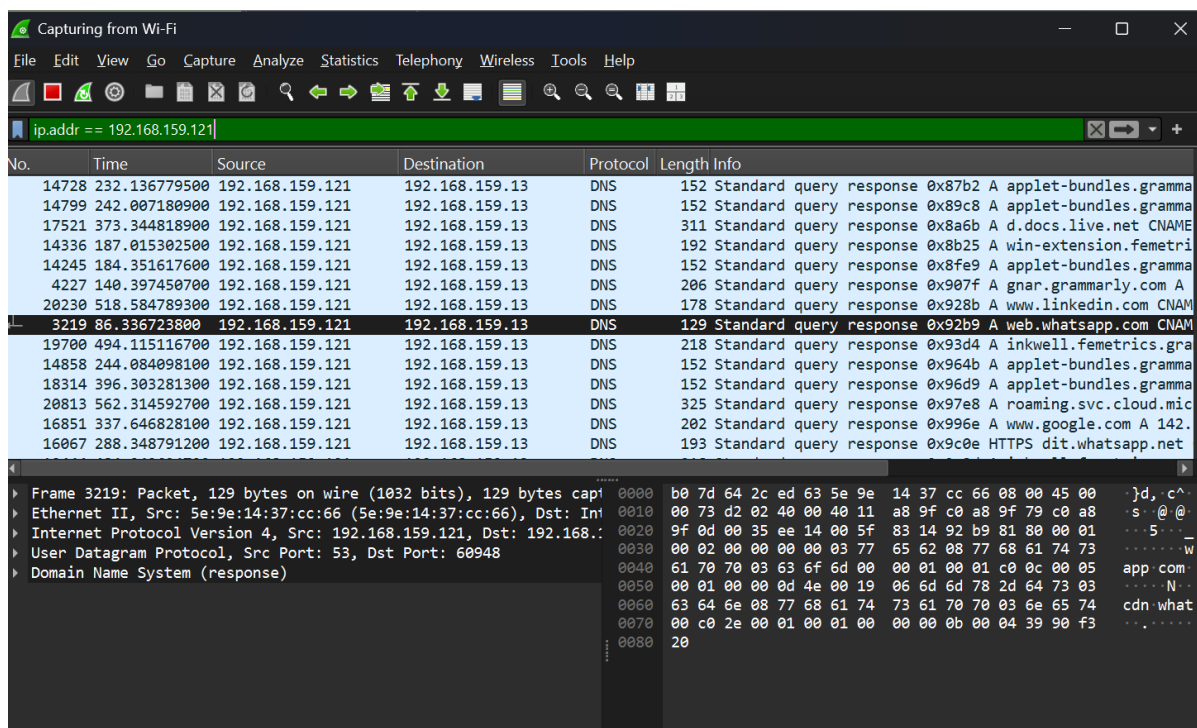


Figure 18: Wireshark capture displaying DNS response traffic originating from the gateway device (192.168.159.121), demonstrating domain name resolution services provided to the local workstation.

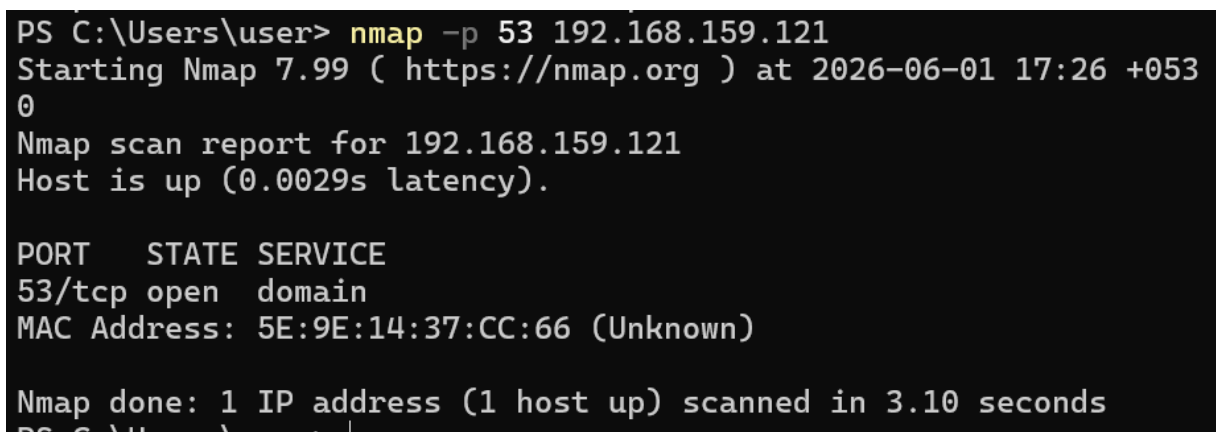


Figure 19: Targeted Nmap port scan performed against the gateway device (192.168.159.121), confirming that TCP port 53 is open and actively providing DNS services.

Wireshark was used to monitor network traffic generated during Nmap port scanning activities. TCP SYN packets were observed as Nmap probed target ports to determine their state. Analysis revealed that open ports responded with SYN-ACK packets, while closed ports returned RST-ACK responses. The packet captures demonstrated the operation of TCP-based reconnaissance techniques and illustrated how network defenders can identify scanning behavior through traffic analysis. This exercise provided insight into the relationship between active reconnaissance tools and the underlying network protocols they utilize.

NETWORK SECURITY ASSESSMENT

```
Nmap done: 1 IP address (1 host up) scanned in 3.10 seconds
PS C:\Users\user> nmap -O 192.168.159.121
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-01 18:26 +0530
Nmap scan report for 192.168.159.121
Host is up (0.0024s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
MAC Address: 5E:9E:14:37:CC:66 (Unknown)
Device type: phone
Running: Google Android 10.X, Linux 4.X
OS CPE: cpe:/o:google:android:10 cpe:/o:linux:linux_kernel:4.14
OS details: Android 10 (Linux 4.14)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 27.77 seconds
```

Figure 20: Nmap operating system fingerprinting results identifying the gateway device as an Android-based system running Android 10 on a Linux 4.14 kernel.

A basic network security assessment was conducted using information collected from packet captures and network reconnaissance activities. Asset identification revealed two active hosts within the local subnet, consisting of a workstation and a gateway device. Service enumeration identified an active DNS service running on TCP port 53 using dnsmasq 2.51. Traffic analysis confirmed normal network operations, including DNS resolution, TCP communications, UDP-based QUIC traffic, and ICMP connectivity testing. No suspicious communication patterns or abnormal network behavior were observed during the assessment period. Security recommendations include maintaining updated firmware, monitoring network services, and performing regular network assessments to identify potential security risks.

Operating system detection was performed using Nmap OS fingerprinting techniques. Analysis indicated that the gateway device (192.168.159.121) was likely running Android 10 on a Linux 4.14 kernel. The device was also observed to provide DNS services through dnsmasq 2.51. The results suggest that the gateway was an Android-based hotspot or mobile tethering device rather than a conventional wireless router. This exercise demonstrated the use of active reconnaissance techniques to identify device characteristics and operating system information within a network environment.

Gateway Assessment

Attribute	Detail
IP Address	192.168.159.121
Device Type	Android mobile hotspot / tethering device
OS	Android 10 (Linux kernel 4.14)
Open Port	TCP 53 (DNS – dnsmasq 2.51)
DNS Service	Provides local DNS resolution via dnsmasq

Local Host Attack Surface Assessment

Nmap service enumeration of the local workstation (localhost) identified the following exposed services:

Port / Protocol	Service	Notes
135 / TCP	Microsoft RPC	Remote Procedure Call – required by many Windows services
445 / TCP	SMB (Microsoft-DS)	File and printer sharing – disable if not required
1433 / TCP	MS SQL Server 2022	Database service – restrict access; apply latest patches
3306 / TCP	MySQL 8.0.43	Database service – restrict to localhost if not serving remote clients

```

PS C:\Users\user> nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-01 18:37 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00065s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
445/tcp    open  microsoft-ds
1433/tcp   open  ms-sql-s
3306/tcp   open  mysql

Nmap done: 1 IP address (1 host up) scanned in 4.41 seconds
PS C:\Users\user> nmap -sV localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-01 18:37 +0530
Nmap scan report for localhost (127.0.0.1)
Host is up (0.00031s latency).
Other addresses for localhost (not scanned): ::1
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
445/tcp    open  microsoft-ds?
1433/tcp   open  ms-sql-s       Microsoft SQL Server 2022 16.00.1180
3306/tcp   open  mysql          MySQL 8.0.43
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.63 seconds

```

Figure 21: Nmap local host assessment identifying open services on the workstation, including Microsoft RPC, SMB, Microsoft SQL Server, and MySQL.

A local host assessment was performed using Nmap to identify exposed services and evaluate the system's attack surface. The scan identified Microsoft RPC (135/TCP), SMB (445/TCP), Microsoft SQL Server 2022 (1433/TCP), and MySQL 8.0.43 (3306/TCP) as active services. Service version detection provided insight into the software running on the host and highlighted potential entry points that could be targeted during a security assessment. The exercise demonstrated the importance of service enumeration and attack surface analysis in identifying systems that may require additional security controls or monitoring.

LOCAL HOST ATTACK SURFACE ASSESSMENT AND SERVICE ENUMERATION

A local host security assessment was conducted using Nmap to identify exposed network services and evaluate the system's attack surface. The assessment involved scanning the localhost to discover open ports and enumerate active services. The results identified Microsoft RPC (135/TCP), Microsoft-DS/SMB (445/TCP), Microsoft SQL Server 2022 (1433/TCP), and MySQL 8.0.43 (3306/TCP) as active services

running on the system. Service version detection was performed to gather detailed information about the exposed applications and their respective versions. The findings demonstrated how network services contribute to a host's attack surface and highlighted the importance of understanding exposed services when performing security assessments. This exercise provided practical experience in service enumeration, attack surface analysis, and vulnerability assessment methodologies commonly used by cybersecurity professionals.

Key Findings

- Identified four active services exposed on the local host.
- Detected Microsoft RPC service running on TCP port 135.
- Detected SMB file-sharing service running on TCP port 445.
- Identified Microsoft SQL Server 2022 running on TCP port 1433.
- Identified MySQL Database Server version 8.0.43 running on TCP port 3306.
- Observed that database services significantly increase the system's attack surface and should be properly secured and monitored.
- Demonstrated the effectiveness of Nmap for host assessment, service discovery, and security analysis.

AUTOMATED PACKET ANALYSIS USING PYTHON

Python and Scapy were utilized to automate the analysis of packet capture files generated through Wireshark. Scripts were developed to process network traffic, classify packets based on protocol types, and generate protocol statistics. Additional analysis was performed to identify the most frequently communicating IP addresses within the capture. The automation reduced the need for manual packet inspection and demonstrated how scripting can enhance network monitoring and security analysis tasks. This activity provided practical experience in cybersecurity automation, packet processing, and traffic pattern identification using Python.

Python scripts were developed to automate analysis of pcap files exported from Wireshark. The improved script (`protocol_counter_improved.py`) incorporates the following enhancements over the original:

- **Modular functions:** `load_packets`, `count_transport_protocols`, `count_ip_addresses`, and `plotting` functions are each isolated for reuse and testability
- **Error handling:** `os.path.exists` check prevents silent crashes if the pcap file is missing
- **Type annotations:** Function signatures use Python type hints for clarity
- **Improved charts:** Annotated bars, colour-coded by protocol, horizontal layout for IP chart, axis gridlines, and clean spines for a professional appearance
- **Console summary:** Structured `print_summary()` outputs totals, percentages, and top-host data in one readable block
- **Configurable constants:** `PCAP_FILE`, `TOP_N_HOSTS`, and `OUTPUT_DIR` are defined at the top for easy adjustment

TRAFFIC VISUALIZATION AND STATISTICAL ANALYSIS

Network traffic statistics obtained through automated packet analysis were visualized using Python and Matplotlib. Protocol distribution charts were generated to illustrate the prevalence of TCP, UDP, and ICMP traffic within the captured dataset. Additional visualizations were created to identify the most active communicating hosts based on packet frequency. These graphical representations improved the interpretability of network activity and demonstrated how security analysts can use data visualization techniques to identify communication trends, network behavior patterns, and potential anomalies more effectively.

Two charts were generated from the packet capture data. Both are saved as high-resolution PNG files (150 dpi) alongside the script.

- **Protocol Distribution (protocol_distribution.png):** Vertical bar chart colour-coded by protocol (TCP, UDP, ICMP, Other) with annotated packet counts
- **Top Communicating Hosts (top_hosts.png):** Horizontal bar chart listing the ten most active IP addresses by packet frequency, sorted descending

Graphical representations improve interpretability and allow analysts to quickly identify dominant protocols and high-frequency communicators that may warrant further investigation

SECURITY RECOMMENDATIONS

- **Firmware and patch management:** Keep the gateway device and all host services updated to mitigate known vulnerabilities
- **Disable unused services:** Review and disable any service not actively required – particularly SMB if file sharing is not needed
- **Restrict database bind addresses:** Configure SQL Server and MySQL to listen on localhost (127.0.0.1) only unless remote access is explicitly required
- **Firewall rules:** Implement host-based firewall rules to limit access to ports 1433 and 3306 to authorised sources
- **Continuous monitoring:** Deploy a SIEM or IDS to alert on anomalous traffic patterns and potential reconnaissance activity
- **Regular assessment cadence:** Repeat network scans and traffic analysis on a scheduled basis to detect configuration drift

RISK ASSESSMENT

The project identified several active services and communication mechanisms within the assessed environment. Based on the analysis, the following risks were observed:

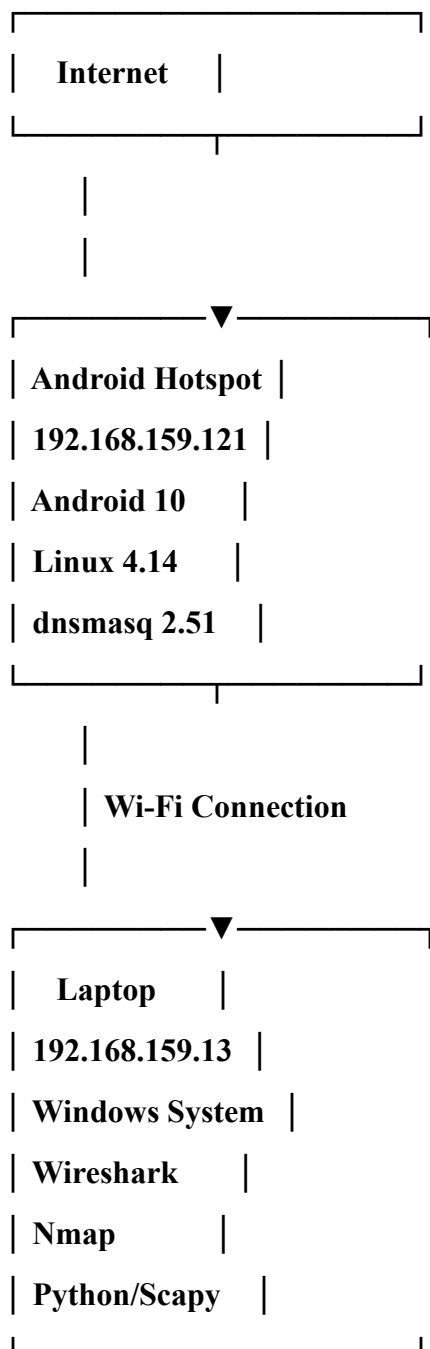
Finding	Risk Level	Description
DNS Service (Port 53)	Low	Required for domain name resolution and normal network operation.
SMB Service (Port 445)	Medium	File-sharing services can be targeted if misconfigured or unpatched.
Microsoft SQL Server (Port 1433)	Medium	Database services may expose sensitive information if not properly secured.

MySQL Service (Port 3306)	Medium	Database services increase the attack surface and require regular security updates.
Exposed Network Services	Medium	Unnecessary or outdated services may increase security risks.

Overall Risk Rating: **Medium**

The identified services are expected within the current environment; however, proper configuration, patch management, and continuous monitoring are recommended to reduce potential security risks.

NETWORK TOPOLOGY DIAGRAM



CONCLUSION

This project successfully demonstrated the practical application of network monitoring, traffic analysis, reconnaissance, service enumeration, attack surface assessment, and cybersecurity automation techniques. Using Wireshark, Nmap, Python, Scapy, and Matplotlib, real network traffic was analyzed and visualized to better understand communication patterns, protocol behavior, and exposed services within the environment.

The project provided valuable hands-on experience in identifying active hosts, analyzing packet-level communications, performing operating system fingerprinting, and assessing network exposure. Additionally, Python-based automation improved the efficiency of traffic analysis and demonstrated the importance of scripting in modern cybersecurity operations.

Overall, the project strengthened practical skills in networking, cybersecurity monitoring, and security assessment methodologies commonly used by security analysts, network administrators, and penetration testers.

FUTURE ENHANCEMENTS

Several improvements can be implemented to further enhance this project:

- Develop a real-time network monitoring dashboard using Flask or Streamlit.
- Implement automated suspicious traffic detection and alerting mechanisms.
- Integrate threat intelligence feeds to identify known malicious IP addresses and domains.
- Perform vulnerability scanning using additional security assessment tools.
- Extend Python automation scripts to generate comprehensive security reports automatically.
- Analyze larger packet capture datasets to improve traffic profiling capabilities.
- Implement machine learning techniques for anomaly detection and network behavior analysis.

These enhancements would transform the project into a more advanced network security monitoring platform and provide deeper insight into cybersecurity operations.